IB DIPLOMA PROGRAMME · PHYSICS

Complete Cheat Sheet

Full Syllabus · HL + SL · Score a 7 · Premium Edition

⚙ Mechanics
🌡 Thermal Physics
🌊 Waves
⚡ Electricity & Magnetism
🪐 Circular Motion
☢ Atomic & Nuclear
🎯 Exam Mastery

🏆 AIM FOR A 7

20 Pages · All Topics Covered
Formulas · Diagrams · Exam Tips

GUIDE

How to Use This Cheat Sheet

📖 HOW TO USE

- **Night before exam:** Read the coloured summary boxes only
- **Topic revision:** Work through each section's formula page with examples
- **Exam strategy:** Use Pages 17–20 for final mindset prep
- **Formula boxes** include variables, units, and when to use each
- ⚠ **Warning boxes** highlight the most common IB mistakes
- 💡 **Exam Insight boxes** tell you exactly what IB examiners love to test
- **Quick Tip bars** at the bottom of each page = fast recall

🎯 LEGEND

- Formula Box – core equation
- 💡 Exam Tip – what IB tests
- ⚠ Warning – common mistakes
- 🔵 Insight – examiner notes
- ✅ Quick Tip – fast recall

VISUAL SYLLABUS ROADMAP

01 Mechanics — Pages 3–5

02 Thermal Physics — Pages 6–7

03 Waves — Pages 8–9

04 Electricity & Magnetism — Pages 10–12

05 Circular Motion & Gravitation — Pages 13–14

06 Atomic & Nuclear Physics — Pages 15–16

07 Exam Mastery — Pages 17–20 · Common mistakes · Exam strategy · Top formulas · Night-before checklist

💡 **IB Physics Assessment Breakdown**

Paper 1: 20% (MCQ, SL 30 Qs / HL 40 Qs) · Paper 2: 40% (short + long structured questions) · Paper 3: 20% (options + data analysis) · Internal Assessment: 20%. Know your mark weightings!

Mechanics – Core Concepts

SCALARS VS VECTORS

Scalar (magnitude only)	Vector (magnitude + direction)
Distance, Speed	Displacement, Velocity
Mass, Time	Acceleration, Force
Energy, Temperature	Momentum, Weight
Electric potential	Electric field strength

Adding vectors: Use components → resolve into x and y, add separately, then recombine using Pythagoras. For 2D: $R = \sqrt{(R_x^2+R_y^2)}$, $\theta = \arctan(R_y/R_x)$

KINEMATIC QUANTITIES

- **Displacement (s)** – change in position from start (m)
- **Velocity (v)** – rate of change of displacement (ms^{-1})
- **Acceleration (a)** – rate of change of velocity (ms^{-2})
- **Average velocity** = total displacement ÷ total time
- **Instantaneous velocity** = gradient of s–t graph at a point
- **Uniform acceleration** = constant change in velocity per unit time

Newton's Three Laws of Motion

1ST LAW – INERTIA

An object remains at rest or moves with constant velocity **unless a net external force acts on it.**

Implication: If $\Sigma F = 0$, then $a = 0$ (equilibrium). Objects in space don't slow down!

2ND LAW – F = MA

The net force on an object equals its mass × acceleration.

$$\Sigma F = ma$$

Also: $F = \Delta p/\Delta t$ (rate of change of momentum)

3RD LAW – ACTION-REACTION

For every action force, there is an **equal and opposite reaction force** acting on a different object.

⚠ The two forces act on DIFFERENT objects — never cancel each other!

Forces Overview

Force	Symbol	Formula	Notes
Weight	W or F_g	$W = mg$	Always vertically downward; $g = 9.81\ ms^{-2}$ on Earth
Normal force	N or F_N	$N = mg\cos\theta$ (on slope)	Always perpendicular to surface
Friction (kinetic)	F_f	$F_f = \mu_k N$	Opposes motion; μ_k = kinetic friction coefficient
Tension	T	—	Acts along rope/string; same throughout massless string
Drag (fluid resistance)	F_D	$F_D \propto v^2$ (at high speeds)	Opposes motion through fluids

Exam Insight – Free Body Diagrams (FBDs)

IB almost always awards 1–2 marks for a correct FBD. Draw arrows from the centre of mass, label each force clearly (e.g. "W = mg downward", "N perpendicular to surface"). Never add "ma" as a force!

Mechanics – Formulas

SUVAT – Equations of Motion (Uniform Acceleration Only)

$v = u + at$

- v: final velocity (ms^{-1})
- u: initial velocity (ms^{-1})
- a: acceleration (ms^{-2})
- t: time (s)

✦ *Use when: displacement not needed*

$s = ut + \frac{1}{2}at^2$

- s: displacement (m)
- u: initial velocity
- a: acceleration
- t: time

✦ *Use when: final velocity not needed*

$v^2 = u^2 + 2as$

- v: final velocity
- u: initial velocity
- a: acceleration
- s: displacement (m)

✦ *Use when: time not needed (most common!)*

$s = \frac{1}{2}(u + v)t$

- s: displacement
- u: initial vel.
- v: final vel.
- t: time

✦ *Use when: acceleration not known*

Energy, Work, Power & Momentum

$W = Fd\cos\theta$

- W: work done (J)
- F: force (N)
- d: distance (m)
- θ: angle between F and d

✦ *θ = 0° → max work; θ = 90° → zero work*

$p = mv$

- p: momentum ($kg\ ms^{-1}$)
- m: mass (kg)
- v: velocity (ms^{-1})

✦ *Vector! Direction = direction of velocity*

$KE = \frac{1}{2}mv^2$

- KE: kinetic energy (J)
- m: mass (kg)
- v: speed (ms^{-1})

✦ *Always positive; v appears squared*

$J = F\Delta t = \Delta p$

- J: impulse ($N\ s = kg\ ms^{-1}$)
- F: average force (N)
- Δt: time interval (s)
- Δp: change in momentum

✦ *Area under F–t graph = impulse*

$GPE = mgh$

- GPE: gravitational PE (J)
- m: mass (kg)
- g: 9.81 ms^{-2}
- h: height (m)

✦ *h relative to reference point*

$P = W/t = Fv$

- P: power ($W = Js^{-1}$)
- W: work done (J)
- t: time (s)
- F: force (N), v: velocity

✦ *P = Fv when F parallel to v*

⚠ **Common Mistake:** SUVAT equations only apply to UNIFORM acceleration. For non-uniform motion, you must use calculus or graphs (area / gradient).

💡 **Efficiency:**
η = useful output power / total input power × 100%. IB loves this — always express as a percentage and check units!

Mechanics – Graphs & Examples

MOTION GRAPHS – KEY RELATIONSHIPS

Graph	Gradient =	Area =
s–t (displacement–time)	velocity	—
v–t (velocity–time)	acceleration	displacement
a–t (acceleration–time)	—	change in velocity
F–t (force–time)	—	impulse
F–x (force–extension)	spring constant k	work done / elastic PE

GRAPH SHAPES TO MEMORISE

- Constant velocity → s–t = straight diagonal, v–t = horizontal
- Uniform acceleration → s–t = parabola, v–t = straight diagonal
- Free fall with drag → v–t starts steep, curves to flat (terminal velocity)
- SHM → s–t = sinusoid; v–t = cosine (90° out of phase)

PROJECTILE MOTION

- **Horizontal:** constant velocity; no acceleration (ignore air resistance)
- **Vertical:** uniform acceleration = g = 9.81 ms^{-2} downward
- Treat H and V components **independently**
- Time of flight is shared between both components
- At max height: $v_y = 0$ (use $v^2 = u^2 - 2gh$)
- Range = $v_x \times$ total time

CONSERVATION LAWS

- **Momentum:** always conserved in closed system (collisions & explosions)
- **Kinetic Energy:** conserved only in elastic collisions
- **Total Energy:** always conserved (KE + PE + heat + …)
- Inelastic collision: momentum conserved; KE lost as heat/sound

Worked Example – Projectile

Q: Ball launched at 20 ms^{-1} at 30° above horizontal. Find range.

- $u_x = 20\cos30° = 17.32$ ms^{-1}
- $u_y = 20\sin30° = 10$ ms^{-1}
- Time to peak: $v_y = 0 \rightarrow t = 10/9.81 = 1.02$ s
- Total time = $2 \times 1.02 = 2.04$ s
- Range = $u_x \times t = 17.32 \times 2.04 \approx$ **35.3 m**

Q: 2 kg block (v=5 ms^{-1}) collides with stationary 3 kg block. They stick. Find v'.

- Conservation of momentum: $m_1u_1 + m_2u_2 = (m_1+m_2)v'$
- $(2)(5) + (3)(0) = (5)v'$
- $10 = 5v' \rightarrow v' =$ **2 ms^{-1}**
- KE before = ½(2)(25) = 25 J
- KE after = ½(5)(4) = 10 J → 15 J lost (inelastic)

⚠ **Top Mistakes:** Forgetting to split projectile into components · Using g = 10 when answer requires 3 sig figs · Confusing distance (scalar) with displacement (vector) · Saying "momentum is conserved in all situations" (needs closed system)

💡 **IB Mark-Getter:**
Always state the principle first ("By conservation of momentum…"), then write the equation, then substitute values. This structure earns method marks even if you get the arithmetic wrong.

Thermal Physics – Concepts & Formulas

KEY CONCEPTS

- **Temperature** – measure of average random kinetic energy of particles
- **Internal Energy** – total KE + PE of all particles in substance
- **Heat (Q)** – thermal energy transferred due to temperature difference
- 0 K = absolute zero; temperature in Kelvin: T(K) = T(°C) + 273
- Specific heat capacity (c) – energy per unit mass per degree change
- Latent heat (L) – energy for phase change at constant temperature

MOLECULAR KINETIC THEORY

- Ideal gas: point particles, no intermolecular forces, elastic collisions
- Pressure due to gas = force from particle collisions on container walls
- Increasing T → increases KE → faster particles → more pressure
- At same T, all gas molecules have same *average* KE (not same speed)
- Mean KE of gas molecule = 3/2 k_B T (HL)

Core Formulas

$Q = mc\Delta T$

Q: heat energy (J) | m: mass (kg)

c: specific heat capacity ($Jkg^{-1}K^{-1}$)

ΔT: temperature change (K or °C)

✦ *Use for heating/cooling WITHOUT phase change*

$PV = nRT$

P: pressure (Pa) | V: volume (m^3)

n: moles of gas (mol) | R: 8.31 $Jmol^{-1}K^{-1}$

T: temperature (K)

✦ *Ideal gas law; T MUST be in Kelvin!*

$Q = mL$

Q: heat energy (J) | m: mass (kg)

L: specific latent heat (Jkg^{-1})

✦ *Use for melting/boiling (temperature constant!)*

$P_1V_1/T_1 = P_2V_2/T_2$

Combined gas law for fixed mass of gas

Boyle's: PV = const (T fixed)

Charles': V/T = const (P fixed)

✦ *Use when gas changes state between two conditions*

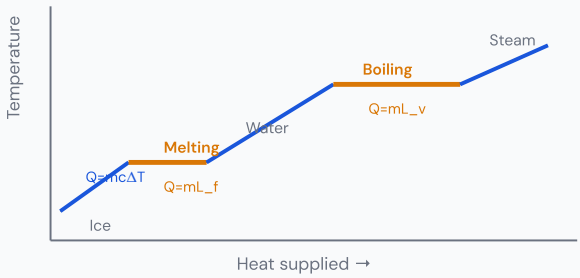
Heating Curve Key Points

Flat sections = phase change (Q = mL); Sloped sections = temperature rising (Q = mcΔT). The gradient of the slope = 1/mc. Steeper slope → lower specific heat capacity.

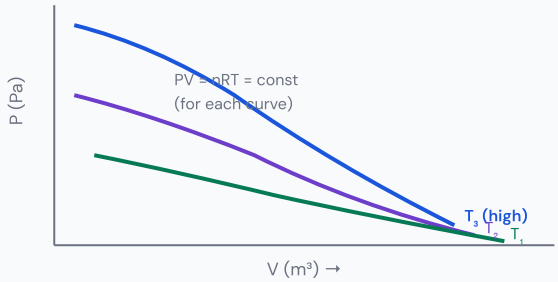
⚠ **Classic Mistake:** Using Celsius instead of Kelvin in PV = nRT or PV/T = const. Always convert °C → K first: T = θ + 273. Losing marks here is easily avoidable!

Thermal Physics – Diagrams & Exam Tips

HEATING CURVE (WATER)



GAS LAWS – P–V DIAGRAMS (ISOTHERMS)



Common IB Thermal Physics Question Types

Question Type	Key Method	Watch Out For
Calculate energy to heat a substance	$Q = mc\Delta T$	Use correct value of c; check units of m (kg not g)
Explain why temperature stays constant during phase change	Energy breaks intermolecular bonds, not increasing KE	Don't say "particles speed up" — they don't!
Find pressure after compression (T constant)	$P_1V_1 = P_2V_2$ (Boyle's)	Volumes must be in same units
Gas at different conditions	$P_1V_1/T_1 = P_2V_2/T_2$	T in Kelvin always; P in Pa or consistent units
Explain molecular model of pressure	Particles hit walls → exert force → P = F/A	Award marks for: "frequent", "elastic", "change in momentum"

💡 **3-Mark Explain Question Template:**
For "explain pressure increases with temperature": (1) Higher T → particles move faster / have higher KE, (2) More frequent/harder collisions with wall, (3) Greater rate of change of momentum → greater force → higher pressure.

✅ QUICK RECALL

- Specific heat water: c = 4200 $Jkg^{-1}K^{-1}$
- Latent heat of fusion water: $L_f \approx 334{,}000\ Jkg^{-1}$
- Latent heat of vaporisation: $L_v \approx 2{,}260{,}000\ Jkg^{-1}$
- R = 8.31 $Jmol^{-1}K^{-1}$ (given in data booklet)

Waves – Core Concepts & Formulas

WAVE PROPERTIES

- **Amplitude (A)** – maximum displacement from equilibrium (m)
- **Wavelength (λ)** – distance for one complete cycle (m)
- **Frequency (f)** – oscillations per second (Hz)
- **Period (T)** – time for one complete cycle; T = 1/f (s)
- **Wave speed (v)** – distance per unit time (ms^{-1})
- **Transverse** – oscillation ⊥ to propagation (light, S-waves)
- **Longitudinal** – oscillation ∥ to propagation (sound, P-waves)

SUPERPOSITION & INTERFERENCE

- **Superposition principle:** resultant displacement = sum of individual displacements
- **Constructive:** path difference = nλ → crests align → amplitude doubles
- **Destructive:** path difference = (n+½)λ → crest meets trough → cancel
- Coherent sources needed: same frequency, constant phase difference
- **Standing wave:** two waves same f, amplitude, speed travelling in opposite directions
- Nodes = zero displacement; Antinodes = maximum displacement

Key Wave Formulas

$v = f\lambda$

`v: wave speed (ms⁻¹)` `f: frequency (Hz)` `λ: wavelength (m)`

✦ *Universal wave equation — always applicable*

$\sin\theta_c = 1/n$ (TIR)

`θ_c: critical angle`

`n: refractive index of denser medium`

✦ *Total internal reflection when θ > θ_c (denser→rarer)*

$n = c/v = \sin\theta_1/\sin\theta_2$

`n: refractive index (dimensionless)`

`c: speed of light (3×10⁸ ms⁻¹)`

`v: speed in medium`

`θ₁: angle in medium 1 (from normal)`

✦ *Snell's Law; angles from the NORMAL not surface*

$\lambda = ax/D$ (double slit)

`λ: wavelength (m)` `a: slit separation (m)`

`x: fringe spacing (m)`

`D: distance to screen (m)`

✦ *Young's double slit experiment*

REFLECTION

- Angle of incidence = angle of reflection
- Both angles measured from the normal
- Phase change of π (180°) on reflection from denser medium

REFRACTION

- Wave bends as it changes speed
- Higher n → slower speed → shorter λ
- Frequency does NOT change on refraction

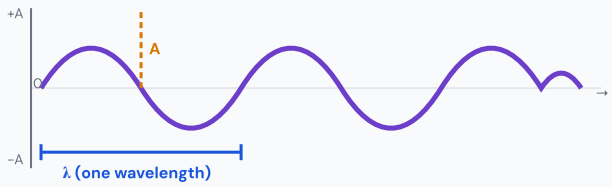
DIFFRACTION

- Waves spread around obstacles / through gaps
- Maximum diffraction: gap width ≈ λ
- Diffraction limited by Rayleigh criterion (HL)

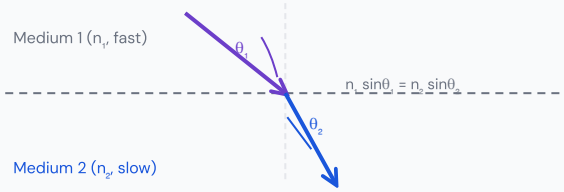
Exam Insight – Doppler Effect:
f_observed = f_source × (v ± v_observer)/(v ∓ v_source). Source approaching → higher frequency (blue shift). Source receding → lower frequency (red shift). IB often asks you to use this for sound AND light (cosmological red shift).

Waves – Diagrams & Sample Problems

TRANSVERSE WAVE DIAGRAM



REFRACTION AT A BOUNDARY



Sample IB Wave Problems

Q: Light travels from glass (n=1.5) to air. Find critical angle.

- $\sin\theta_c = 1/n = 1/1.5 = 0.667$
- $\theta_c = \arcsin(0.667) =$ **41.8°**
- If angle of incidence > 41.8°: Total Internal Reflection

Q: Young's double slit: a=0.5mm, D=2m, x=2.4mm. Find λ.

- $\lambda = ax/D = (0.5\times10^{-3})(2.4\times10^{-3})/2$
- $\lambda = 1.2\times10^{-6}/2 =$ **6.0×10^{-7} m = 600 nm**
- (This is orange/red visible light ✓)

STANDING WAVES – HARMONICS

- String fixed both ends: $\lambda_n = 2L/n$; $f_n = nv/2L$
- Pipe open both ends: same as string formula
- Pipe closed one end: $\lambda_n = 4L/n$ (odd harmonics only)
- Fundamental = 1st harmonic (n=1)
- At nodes: always displacement = 0

Polarisation (HL):

Only transverse waves can be polarised. Malus's Law: $I = I_0\cos^2\theta$ (intensity after polariser). If θ = 90°, I = 0. IB loves questions where you rotate a polariser and calculate intensity changes.

Electricity – Core Formulas

Fundamental Electrical Formulas

$V = IR$

V: potential difference (V) | I: current (A) | R: resistance (Ω)

✦ *Ohm's Law; only for ohmic conductors (constant R)*

$E = VQ = Pt$

E: energy (J) | V: voltage (V) | Q: charge (C) | P: power (W); t: time (s)

✦ *Energy transferred by charge moving through pd*

$P = IV = I^2R = V^2/R$

P: power (W) | I: current (A) | V: voltage (V) | R: resistance (Ω)

✦ *Choose formula based on what's given*

$\varepsilon = V + Ir$ (EMF)

ε: EMF of cell (V) | V: terminal pd (V) | I: current (A) | r: internal resistance (Ω)

✦ *Terminal voltage drops when current drawn is high*

$Q = It$

Q: charge (C) | I: current (A) | t: time (s)

✦ *Current = charge per unit time; I = ΔQ/Δt*

$R = \rho L/A$

R: resistance (Ω) | ρ: resistivity (Ωm) | L: length (m) | A: cross-sectional area (m²)

✦ *Material property; longer/thinner wire = more R*

KIRCHHOFF'S LAWS

- **KCL (Junction rule):** $\Sigma I_{in} = \Sigma I_{out}$ at any junction (conservation of charge)
- **KVL (Loop rule):** Sum of EMFs = Sum of potential drops around any closed loop
- Apply systematically: label currents, choose loop direction, be consistent with signs

I–V CHARACTERISTICS

Component	I–V Graph Shape
Ohmic resistor	Straight line through origin
Filament bulb	Curve (steeper at high V — R increases)
Diode	Flat (reverse bias), then steep rise (forward)
NTC Thermistor	Curve (R decreases as T increases)

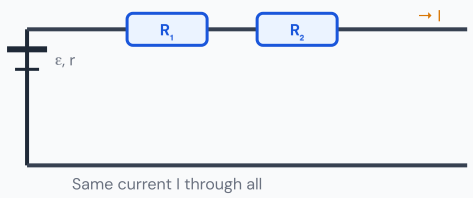
Exam Insight – Potential Divider:
$V_{out} = V_{in} \times R_2/(R_1+R_2)$. The output is taken across R_2. IB loves using this with thermistors and LDRs in sensor circuits. As temperature rises → thermistor R decreases → V_{out} drops.

Electricity – Circuits

Series vs Parallel Circuits

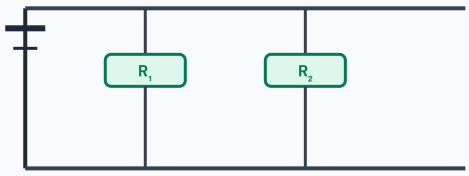
Property	Series Circuit	Parallel Circuit
Current	Same through all components: $I_1 = I_2 = I_3$	Splits: $I_{total} = I_1 + I_2 + I_3$
Voltage	Splits: $V_{total} = V_1 + V_2 + V_3$	Same across all branches: $V_1 = V_2 = V_3$
Resistance	$R_{total} = R_1 + R_2 + R_3$ (adds up)	$1/R_{total} = 1/R_1 + 1/R_2 + 1/R_3$
If one fails	All fail (circuit breaks)	Others continue working
Real use	Fairy lights, simple circuits	Household wiring, parallel branches

CIRCUIT DIAGRAM – SERIES



Same current I through all

CIRCUIT DIAGRAM – PARALLEL



Same voltage across each branch

Worked Example – Mixed Circuit

Q: R_1=6Ω and R_2=3Ω in parallel, then in series with R_3=4Ω. EMF = 12V, r = 0. Find I through R_3.

Step 1: Parallel combination

$1/R_p = 1/6 + 1/3 = 1/6 + 2/6 = 3/6$

$R_p = 2\ \Omega$

Step 2: Total resistance

$R_{total} = R_p + R_3 = 2 + 4 = 6\ \Omega$

Step 3: Total current

$I = \varepsilon/R = 12/6 =$ **2 A**

(through R_3 and the parallel pair)

⚠ **Mistake:** Adding resistors in parallel using $R_1 + R_2$. WRONG. Use $1/R_{total} = 1/R_1 + 1/R_2$. For two resistors only: $R_{total} = R_1R_2/(R_1+R_2)$

💡 **Ammeters & Voltmeters:**

Ideal ammeter has R=0 (placed in series). Ideal voltmeter has R=∞ (placed in parallel). Real meters affect circuit — IB sometimes asks you to account for this.

Magnetism – Fields & Forces

Magnetic Force Formulas

$F = BIL \sin\theta$

- F: magnetic force (N)
- B: magnetic field strength (T)
- I: current (A)
- L: length of conductor in field (m)
- θ: angle between I and B

✦ *Max force when θ=90°; zero when θ=0° (parallel)*

$F = qvB \sin\theta$

- F: force on moving charge (N)
- q: charge (C)
- v: velocity of charge (ms^{-1})
- B: field strength (T)
- θ: angle between v and B

✦ *Charge moving ⊥ to B → circular motion (Lorentz)*

MAGNETIC FIELD DIRECTIONS (RIGHT-HAND RULES)

- **Straight wire:** Curl fingers in direction of field; thumb points in direction of conventional current
- **Force on wire (F=BIL):** Point fingers in direction of I, curl toward B; thumb = direction of force
- **Solenoid:** Curl fingers in direction of current in coils; thumb points to North pole
- For **negative charge**: reverse the direction of force

FIELD LINE DIAGRAMS



Electromagnetic Induction (Faraday & Lenz)

FARADAY'S LAW

$$\varepsilon = -\Delta\Phi/\Delta t = -N\,\Delta\Phi/\Delta t$$

- ε: induced EMF (V); Φ: magnetic flux (Wb = Tm^2)
- Flux Φ = BAcosθ (B × Area perpendicular)
- Change flux faster → larger induced EMF
- N turns: multiply EMF by N (solenoid)

LENZ'S LAW

- Induced current opposes the change that caused it
- Negative sign in Faraday's law shows this opposition
- Example: magnet entering coil → induced B opposes entering field
- Acts to conserve energy (no perpetual motion!)

Circular Motion in Magnetic Field:
F = qvB provides centripetal force: $qvB = mv^2/r \rightarrow r = mv/qB$. Heavier particle or higher speed = larger radius. This is the basis of mass spectrometers and particle accelerators — classic IB HL question!

Circular Motion & Gravitation

Circular Motion Formulas

$$v = \omega r$$

- v: linear (tangential) speed (ms^{-1})
- ω: angular velocity ($rads^{-1}$)
- r: radius of circular path (m)

✦ *$\omega = 2\pi f = 2\pi/T$; T = period (s)*

$$F_c = mv^2/r = m\omega^2 r$$

- F_c: centripetal force (N)
- m: mass (kg)
- v: speed (ms^{-1})
- r: radius (m)

✦ *Not a new force — provided by gravity, tension, normal force, friction, etc.*

$$a_c = v^2/r = \omega^2 r$$

- a_c: centripetal acceleration (ms^{-2})
- v: speed (ms^{-1})
- r: radius (m)
- ω: angular velocity ($rads^{-1}$)

✦ *Always directed toward the CENTRE of circle*

KEY CONCEPTS

- Speed constant in uniform circular motion, but velocity changes (direction changes)
- Acceleration is NOT zero — direction always changes
- Centripetal force points inward; velocity is tangential (outward)
- No "centrifugal force" — it's a fictitious force in rotating frames

Common Circular Motion Scenarios

Scenario	What provides F_c?	Key equation
Planet orbiting Sun	Gravitational force	$GMm/r^2 = mv^2/r$
Car on flat road	Friction	$f = mv^2/r$
Car on banked road	Normal force (horizontal component)	$N\sin\theta = mv^2/r$
Ball on string (horizontal)	Tension (horizontal component)	$T\sin\theta = mv^2/r$
Vertical circle (top)	$mg + N = mv^2/r$	Min speed: $v = \sqrt{(gr)}$ when N=0
Satellite in orbit	Gravity	$GM/r^2 = v^2/r \rightarrow v = \sqrt{(GM/r)}$

Vertical Circles – Critical Point at the Top:
At the minimum speed for circular motion, normal force N = 0. Then: $mg = mv^2/r \rightarrow v_{min} = \sqrt{(gr)}$. Below this speed, the object falls away from the circular path before reaching the top.

⚠ **Common Mistake:** Thinking centripetal force is a separate real force. It is always provided by existing forces (gravity, tension, friction). Never draw F_c as a separate arrow on a free body diagram!

Gravitation & Orbital Motion

$F = GMm/r^2$

F: gravitational force (N)

G: 6.67×10^{-11} Nm^2kg^{-2} (given)

M, m: masses (kg)

r: distance between centres (m)

✦ *Newton's Law of Gravitation — always attractive*

$g = GM/r^2$

g: gravitational field strength (Nkg^{-1} = ms^{-2})

M: mass of planet/body (kg)

r: distance from centre (m)

✦ *Field strength decreases with r^2 (inverse square law)*

$T^2 = (4\pi^2/GM)\ r^3$

T: orbital period (s)

r: orbital radius (m)

M: mass of central body (kg)

G: gravitational constant

✦ *Kepler's 3rd Law: $T^2 \propto r^3$*

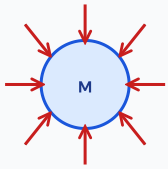
$E_p = -GMm/r$

E_p: gravitational PE (J) — always negative!

Zero at infinity; becomes more negative as r decreases

✦ *Total orbital energy = KE + E_p = $-GMm/2r$*

GRAVITATIONAL FIELD DIAGRAM



Lines point toward mass (inward)

ORBITAL MECHANICS SUMMARY

- **Orbital speed:** $v = \sqrt{(GM/r)}$ — faster closer to planet
- **Geostationary orbit:** T = 24h, r ≈ 42,000 km from Earth centre; stays above same point
- **LEO (Low Earth Orbit):** r ≈ 6,400–8,000 km; T ≈ 90 min
- **Escape velocity:** $v_e = \sqrt{(2GM/r)}$ — approx 11.2 kms^{-1} from Earth surface
- Larger orbit → slower speed, longer period, less total energy (more negative)

💡 **Deriving Kepler's 3rd Law (worth learning!):** Set gravitational force = centripetal: $GMm/r^2 = mv^2/r$ → $v^2 = GM/r$. Then $v = 2\pi r/T$ → $(2\pi r/T)^2 = GM/r$ → $T^2 = 4\pi^2r^3/GM$. IB may ask you to show this derivation — it's 3–4 marks.

⚠ **Mistake:** Using r = height above surface. Always use distance from the CENTRE of the Earth: $r = R_{Earth} + h = 6.4\times10^6 + h$ (m). This mistake loses marks in almost every orbital question.

Atomic & Nuclear Physics – Core Concepts

ATOMIC STRUCTURE

- Atom = nucleus (protons + neutrons) + electron shells
- **Proton number (Z)** = atomic number = number of protons
- **Nucleon number (A)** = mass number = protons + neutrons
- **Neutron number (N)** = A − Z
- Isotopes: same Z, different A (different neutron count)
- Energy levels: electrons occupy discrete energy levels; can jump by absorbing/emitting photons

RADIOACTIVE DECAY TYPES

Type	Particle	Penetration	Change
α	$^{4}_{2}He$	Low (paper)	Z−2, A−4
β^-	e^- + antineutrino	Med (Al)	Z+1, A same
γ	Photon (EM)	High (Pb)	Z same, A same

Key Nuclear Formulas

$E = hf = hc/\lambda$

- E: photon energy (J)
- h: 6.63×10^{-34} Js (Planck's constant)
- f: frequency (Hz)
- c: 3×10^{8} ms^{-1}

✦ *Energy of a photon; use when light = photons*

$N = N_0\ e^{(-\lambda t)}$

- N: number of nuclei at time t
- N_0: initial number of nuclei
- λ: decay constant (s^{-1})
- t: time (s)

✦ *Exponential decay; also applies to activity $A = A_0 e^{(-\lambda t)}$*

$E = mc^2$

- E: energy (J)
- m: mass converted to energy (kg)
- c: 3×10^{8} ms^{-1}

✦ *Mass-energy equivalence; use for nuclear reactions, binding energy*

$t_{1/2} = \ln 2/\lambda = 0.693/\lambda$

- $t_{1/2}$: half-life (s, min, yr — consistent with λ)
- λ: decay constant (s^{-1})

✦ *After n half-lives: $N = N_0 \times (1/2)^n$*

PHOTOELECTRIC EFFECT (HL)

$$hf = \varphi + \tfrac{1}{2}mv^2_{max}$$

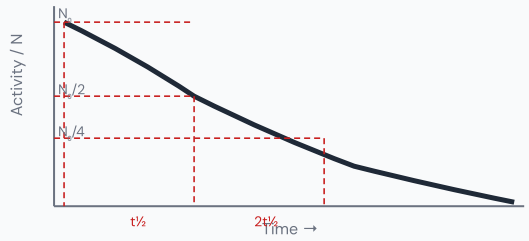
- φ: work function (min energy to eject electron)
- Threshold frequency: $f_0 = \varphi/h$ (minimum f for emission)
- Increasing intensity → more photons → more electrons (not more KE)
- Increasing frequency → higher max KE of ejected electrons

Nuclear Notation:

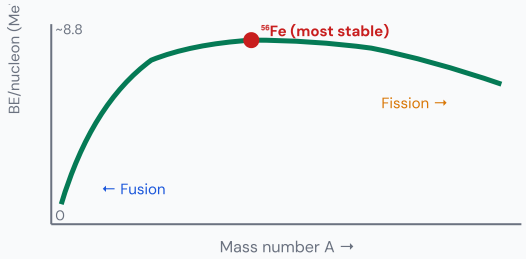
$^{A}_{Z}X$ — top number A (nucleons), bottom Z (protons). In any nuclear equation: ΣA must balance AND ΣZ must balance. Check both sides! E.g.: $^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$ (alpha decay — verify: 238=234+4 ✓, 92=90+2 ✓)

Atomic & Nuclear – Diagrams & Details

RADIOACTIVE DECAY CURVE



BINDING ENERGY PER NUCLEON



Nuclear Fission & Fusion

NUCLEAR FISSION

- Heavy nucleus splits into 2 lighter nuclei + neutrons + energy
- Example: $^{235}U + n \rightarrow {}^{141}Ba + {}^{92}Kr + 3n$ + energy
- Energy released = mass defect × c^2; both daughters more stable (higher BE/nucleon)
- Chain reaction if neutrons captured by other ^{235}U nuclei
- Critical mass = minimum mass for self-sustaining chain reaction

NUCLEAR FUSION

- Light nuclei combine to form heavier nucleus + energy
- Example: ${}^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + {}^{1}_{0}n$ + 17.6 MeV
- Product has higher BE/nucleon → more stable → energy released
- Requires very high temperature (10^7–10^8 K) to overcome Coulomb barrier
- Power source of stars; potential future clean energy

Mass Defect & Binding Energy:
Mass defect Δm = (sum of separate nucleon masses) − (actual nuclear mass). Binding energy = Δm × c^2. Higher binding energy per nucleon = more stable nucleus. Energy released in reaction = binding energy of products − binding energy of reactants.

Common Mistakes – Avoid These!

Mechanics

⚠ **1.** Using g = 10 instead of 9.81 ms^{-2} when the question requires 3 sig figs

⚠ **2.** Confusing distance (scalar) with displacement (vector) — direction matters!

⚠ **3.** SUVAT applied to non-uniform acceleration (e.g., a changing, or on a curve)

⚠ **4.** Drawing "ma" or "centripetal force" on a free body diagram as if they're new forces

⚠ **5.** Assuming elastic collision when problem says "stick together" (perfectly inelastic!)

Thermal Physics

⚠ **6.** Using Celsius instead of Kelvin in ideal gas equations

⚠ **7.** Saying "particles move faster during melting" — temperature/KE stays constant during phase change

⚠ **8.** Confusing specific heat capacity (c) with specific latent heat (L) — different formulas!

Waves & Optics

⚠ **9.** Measuring angles for Snell's Law from the surface instead of the normal

⚠ **10.** Thinking frequency changes during refraction — only speed and wavelength change!

Electricity

⚠ **11.** Adding parallel resistors directly: $R_{total} \neq R_1 + R_2$. Use $1/R_{total} = 1/R_1 + 1/R_2$

⚠ **12.** Forgetting internal resistance when finding terminal pd: $V = \varepsilon - Ir$, not $V = \varepsilon$

Circular Motion & Nuclear

⚠ **13.** Using height above surface instead of distance from Earth's centre in orbital mechanics

⚠ **14.** Not balancing both A and Z in nuclear decay equations (check both sides!)

⚠ **15.** Thinking intensity (brightness) changes photon energy in photoelectric effect — only frequency does

💡 **Mark-Saving Rule:**
IB examiners follow mark schemes strictly. Even if your final answer is wrong due to arithmetic, you can still earn method marks (ECF — error carried forward). Always show all working and state the formula you're using before substituting values.

Exam Strategy & Time Management

APPROACHING IB PHYSICS QUESTIONS

- **Read carefully:** Identify what's given and what's being asked
- **Define variables:** Write out knowns and unknowns before picking formula
- **State the principle:** "By Newton's 2nd Law..." / "By conservation of momentum..."
- **Write formula:** Show the equation before substituting
- **Substitute with units:** Always include units at each step
- **Check magnitude:** Does the answer make physical sense?
- **Round correctly:** Match sig figs to least precise given value (usually 3)

COMMAND TERMS – WHAT IB MEANS

Term	What it means
State	Give without justification (1–2 marks)
Define	Give precise meaning, include units if relevant
Explain	Link cause and effect, use physics principles
Describe	Give key features, no explanation needed
Deduce	Show logical reasoning from given info
Estimate	Approximate to 1 sig fig is acceptable
Sketch	Rough diagram, must show key features
Discuss	Give arguments for/against, consider all aspects

Time Management per Paper

Paper	Duration	Content	Strategy
Paper 1	SL: 45min / HL: 1hr	MCQ (no penalty for wrong)	Answer ALL. Eliminate 2 wrong answers first. Guess if unsure — no penalty!
Paper 2	SL: 75min / HL: 2.25hr	Short + extended response	~1.5 min per mark. Tackle known topics first. Leave blanks for second pass.
Paper 3	SL: 45min / HL: 1.25hr	Data + options	Data analysis is first — do it carefully. Options: do the one you know best.

PAPER 2 EXPLAIN QUESTIONS – STRUCTURE

- **3-mark explains follow a pattern:** (1) State the physics principle, (2) Apply to the situation, (3) Draw conclusion / state outcome
- Use precise vocabulary: "collisions", "rate of change of momentum", "work done against resistance"
- Avoid: "faster", "more energy" — be specific about why
- For graph questions: always describe gradient and area as separate features

GAINING EASY MARKS

- Always draw and label a free body diagram when asked
- Always include units on every answer
- In "show that" questions, work to MORE decimal places than the given answer
- For uncertainty calculations, express to 1 sig fig only
- In Paper 3 data analysis: always draw a best-fit line before calculating gradient
- State conservation laws explicitly — they're worth marks

Top Formulas – Master Reference

Mechanics

$v = u + at$

$s = ut + \frac{1}{2}at^2$

$v^2 = u^2 + 2as$

$F = ma$

$W = Fd\cos\theta$

$KE = \frac{1}{2}mv^2$

$GPE = mgh$

$p = mv$

$J = F\Delta t = \Delta p$

$P = Fv = W/t$

Thermal + Waves

$Q = mc\Delta T$

$Q = mL$

$PV = nRT$

$P_1V_1/T_1 = P_2V_2/T_2$

$v = f\lambda$

$n = c/v = \sin\theta_1/\sin\theta_2$

$\sin\theta_c = 1/n$

$\lambda = ax/D$

$I = I_0\cos^2\theta$ (Malus)

E&M + Circular + Nuclear

$V = IR$

$P = IV = I^2R$

$\varepsilon = V + Ir$

$F = BIL\sin\theta$

$F = qvB\sin\theta$

$F = mv^2/r = m\omega^2 r$

$F = GMm/r^2$

$T^2 = 4\pi^2r^3/GM$

$E = hf$

$E = mc^2$

$N = N_0e^{(-\lambda t)}$

$t_{\frac{1}{2}} = \ln 2/\lambda$

KEY CONSTANTS (DATA BOOKLET)

Speed of light	$c = 3\times10^8\ ms^{-1}$
Planck's constant	$h = 6.63\times10^{-34}\ Js$
Gravitational constant	$G = 6.67\times10^{-11}$
Gas constant	$R = 8.31\ Jmol^{-1}K^{-1}$
Electron charge	$e = 1.60\times10^{-19}\ C$
Electron mass	$m_e = 9.11\times10^{-31}\ kg$

Data Booklet Strategy:
IB gives you the data booklet in every exam. Practise finding formulas quickly — don't waste time searching. Know which section each formula is in. Many students don't realise some HL-only formulas are in there too. Highlight your most-used formulas on the first practice exam.

Final Revision – Night Before the Exam

Ultra-Condensed Key Points

MECHANICS ⚙

- SUVAT: uniform acceleration only; check 5 variables, pick 3
- Newton 3: equal/opposite forces on DIFFERENT objects
- Impulse = area under F–t graph = Δp
- Energy always conserved; KE only in elastic collisions
- Gradient v–t = a; area v–t = displacement

THERMAL 🌡

- T must be in Kelvin for all gas law equations
- Phase change: Q=mL (T constant, bonds break/form)
- Ideal gas: PV=nRT; molecules elastic, point-like
- Higher T → faster particles → more pressure

WAVES 🌊

- v = fλ; refraction changes v and λ, NOT f
- Snell's Law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$; angles from NORMAL
- Constructive: path diff = nλ; Destructive: (n+½)λ
- Standing waves: nodes (zero disp.) + antinodes

Night Before Checklist

- [ ] I know all SUVAT equations from memory
- [ ] I can draw a correct free body diagram
- [ ] I know Newton's 3 Laws in my own words
- [ ] I convert °C to K before using gas laws
- [ ] I can apply Q=mcΔT and Q=mL correctly
- [ ] I know v=fλ and when to use Snell's Law
- [ ] I can add series and parallel resistances
- [ ] I know the right-hand rule for magnetic forces
- [ ] I use r = R_Earth + h in orbital problems
- [ ] I can balance nuclear equations (check A and Z)
- [ ] I know $N = N_0 e^{(-\lambda t)}$ and the half-life formula
- [ ] I understand what centripetal force IS (not a new force)
- [ ] I can apply conservation of momentum to collisions
- [ ] I know the IB command terms (state, explain, deduce…)
- [ ] I've reviewed my marked practice papers
- [ ] I have my data booklet ready for tomorrow

🏆 Final Mindset

✓ Show all working — method marks save you
✓ State conservation laws before equations
✓ Never leave MCQ blank — no penalty
✓ Check units at every step
✓ Answer in terms of the mark count
✓ Sleep well tonight — you've got this!